

Reporting survival analysis results with the `{gtsummary}` and `{ggsurvfit}` packages

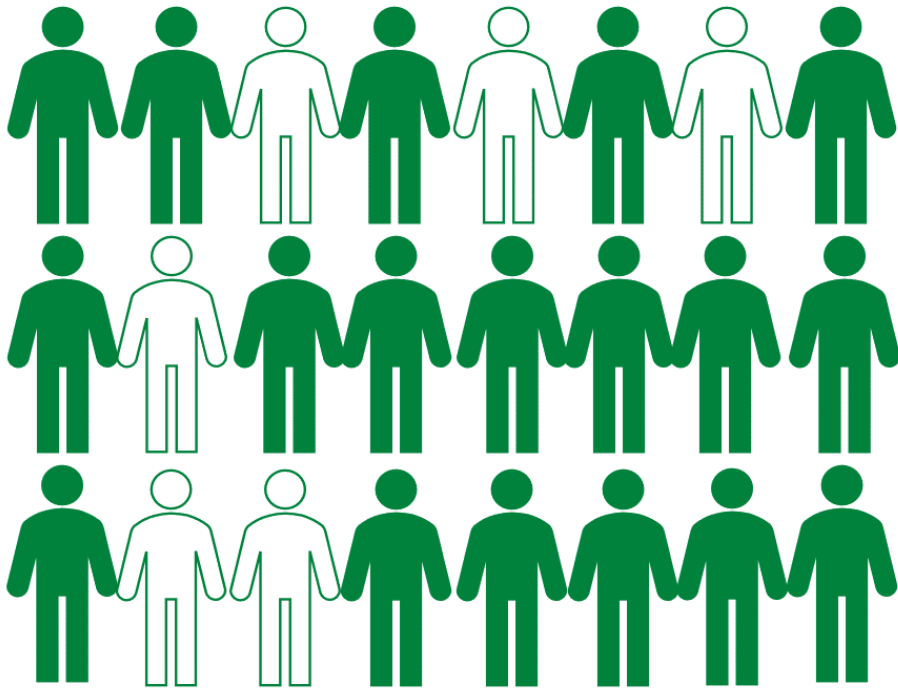
NYR Conference 2024

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Time-to-event endpoints are common in biomedical research

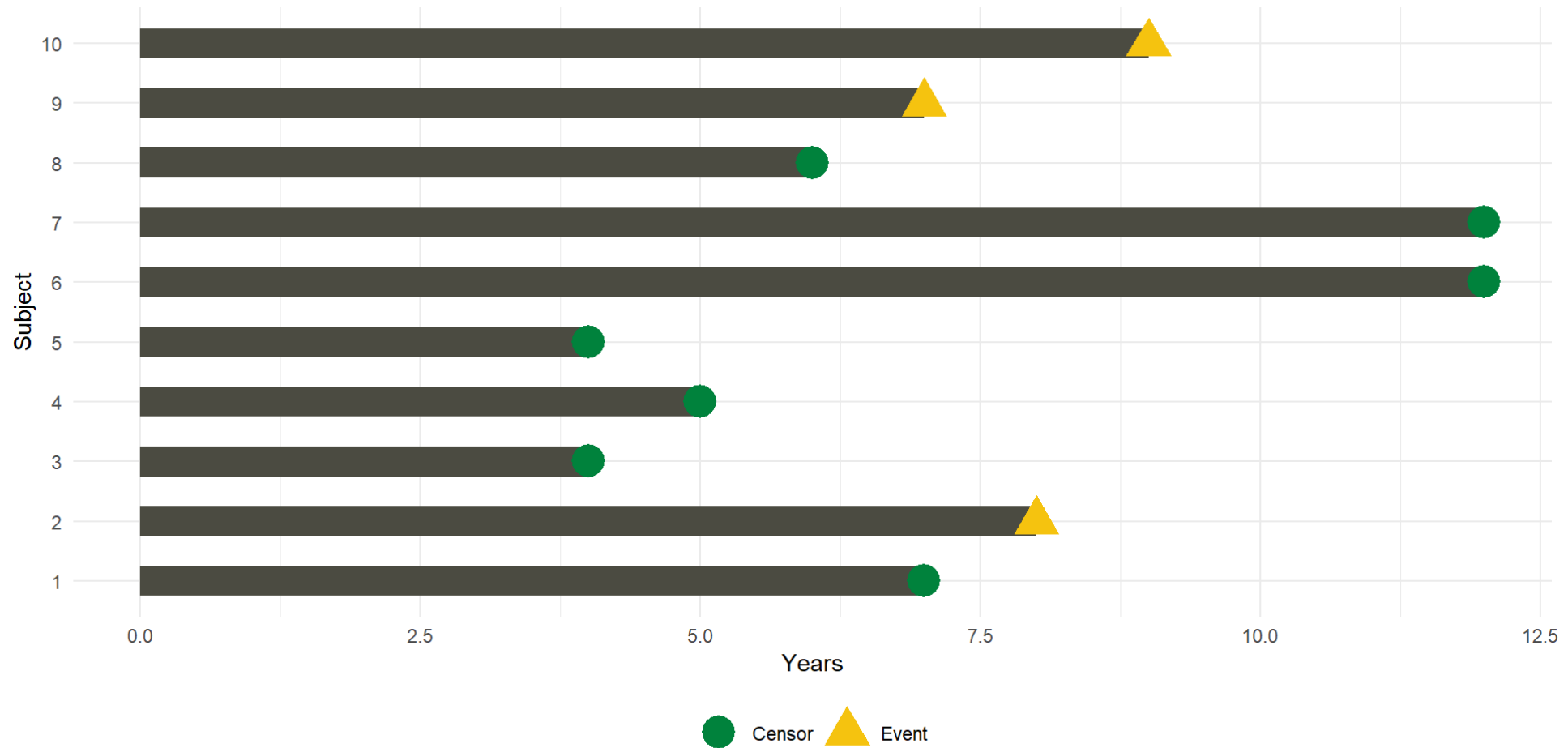


Common research questions relate to times and probabilities



- What is the probability of remaining event-free for a certain number of years?
- What is the average amount of event-free time?

Survival analysis methods needed when follow-up time varies



The {survival} package is the basis of the survival ecosystem in R



- Began development in 1985
- Total of 11.9M downloads
- Active development ongoing
- Many detailed vignettes covering both the basics and advanced topics
- Includes the essential methods

```
1 install.packages("survival")  
2 library(survival)
```

Example dataset

The `colon` data come from a clinical trial of adjuvant therapy for colon cancer, and are included in the `{survival}` package.

Variables of interest:

Variable	Variable description	Levels
rx	Treatment assignment	Obs(ervation), Lev(amisole), Lev(amisole)+5-FU
sex	Sex	1=male, 0=female
years	Years to status	Continuous years
status	Censoring status	0=censored, 1=recurrence

See appendix slide for code to generate altered example data `colonr`.

Estimate median event-free time

Use `survfit()` to create the survival curves from which the median recurrence-free years and 95% confidence interval are obtained:

```
1 survfit(Surv(years, status) ~ 1, data = colonr)
```

```
Call: survfit(formula = Surv(years, status) ~ 1, data = colonr)
```

	n	events	median	0.95LCL	0.95UCL
[1,]	929	468	5.52	4.01	NA

Estimate x-year event-free probability

Run `summary()` on the `survfit()` object to estimate the 5-year recurrence-free probability and 95% confidence interval:

```
1 summary(survfit(Surv(years, status) ~ 1, data = colonr), times = 5)
```

```
Call: survfit(formula = Surv(years, status) ~ 1, data = colonr)
```

time	n.risk	n.event	survival	std.err	lower	95% CI upper	95% CI
5	437	451	0.508	0.0166	0.476	0.541	

Fit Cox regression models

Use `coxph()` to fit univariate and multivariable Cox regression models:

```
1 coxph(Surv(years, status) ~ rx, data = colonr)
```

Call:

```
coxph(formula = Surv(years, status) ~ rx, data = colonr)
```

	coef	exp(coef)	se(coef)	z	p
rxLev	-0.01512	0.98499	0.10708	-0.141	0.888
rxLev+5FU	-0.51209	0.59924	0.11863	-4.317	1.58e-05

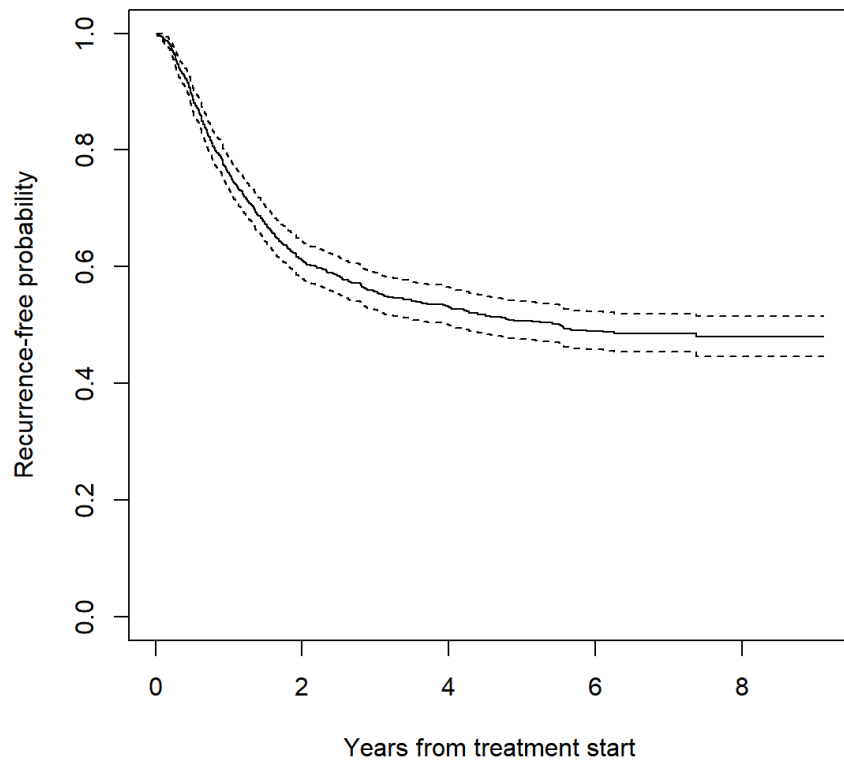
Likelihood ratio test=24.34 on 2 df, p=5.175e-06

n= 929, number of events= 468

Plot Kaplan-Meier curves

Run `plot()` on the `survfit()` object to generate the Kaplan-Meier plot:

```
1 plot(survfit(Surv(years, status) ~ 1, data = colonr),  
2     xlab = "Years from treatment start",  
3     ylab = "Recurrence-free probability")
```



Done! That's everything we need!

But what if we need to put our results in a reproducible and publication-ready report?



Generates
results

+



Puts results
in tables

+



Creates
plots

=



```
1 install.packages(c("gtsummary", "ggsurvfit"))
2 library(gtsummary)
3 library(ggsurvfit)
```

Survival tables with {gtsummary}

Use `tbl_survfit()` to create:

- Tables of median event-free time
- Tables of x-time event-free probability

Use `tbl_uvregression()` to create:

- Tables of univariate Cox regression results

Use `tbl_regression()` to create:

- Tables of multivariable Cox regression results

Tables are highly customizable, see <https://www.danielsjoberg.com/gtsummary/> for details and vignettes to get started

Table of median event-free time

Use `tbl_survfit()` with the `probs = 0.5` argument:

```
1 list(  
2   survfit(  
3     Surv(years, status) ~ 1,  
4     data = colonr),  
5   survfit(  
6     Surv(years, status) ~ rx,  
7     data = colonr)  
8 ) |>  
9 tbl_survfit(  
10   probs = 0.5,  
11   label_header =  
12     "Median (95% CI)"  
13 )
```

Characteristic	Median (95% CI)
Overall	5.5 (4.0, —)
Treatment	
Obs	3.4 (2.2, 5.6)
Lev	3.2 (2.2, 5.7)
Lev+5FU	— (—, —)

- Highly customizable
- Can set labels, titles, subtitles, footnotes, and more

Table of x-year event-free probability

Use `tbl_survfit()` with the `times = 5` argument:

```
1 list(  
2   survfit(  
3     Surv(years, status) ~ 1,  
4     data = colonr),  
5   survfit(  
6     Surv(years, status) ~ rx,  
7     data = colonr)  
8 ) |>  
9   tbl_survfit(  
10    times = 5,  
11    label_header =  
12      "{time}-year RFS (95% CI)"  
13  )
```

Characteristic	5-year RFS (95% CI)
Overall	51% (48%, 54%)
Treatment	
Obs	45% (40%, 51%)
Lev	46% (41%, 52%)
Lev+5FU	62% (56%, 67%)

- Note the use of {glue} syntax in the label header to dynamically code the column header

Inline results reporting

Save the `tbl_survfit()` result to an object (here, `tab1`) so that table statistics can be extracted for use inline with the `inline_text()` function.

The inline code:

```
The median (95% CI) recurrence-free time for the Levamisole treatment group is `r  
inline_text(tab1, variable = rx, level = 'Lev', prob = 0.5)`.
```

Will show as:

```
The median (95% CI) recurrence-free time for the Levamisole treatment group is 3.2  
(2.2, 5.7).
```


Univariate Cox regression table

Use `tbl_uvregression()`:

```
1 colonr |>
2   tbl_uvregression(
3     method = coxph,
4     y = Surv(years, status),
5     include = c(rx, sex),
6     exponentiate = T
7   ) |>
8   add_global_p()
```

Characteristic	N	HR ¹	95% CI ¹	p-value
Treatment	929			<0.001
Obs		—	—	
Lev		0.98	0.80, 1.21	
Lev+5FU		0.60	0.47, 0.76	
Sex	929			0.4
Female		—	—	
Male		0.92	0.77, 1.10	
¹ HR = Hazard Ratio, CI = Confidence Interval				

- Fits models for all variables listed in `include =`
- `exponentiate = T` to obtain hazard ratios
- Add global p-values

Multivariable Cox regression table

Pass fitted model results directly to `tbl_regression()`:

```
1 coxph(  
2   Surv(years, status) ~  
3     rx + sex,  
4   data = colonr  
5 ) |>  
6 tbl_regression(  
7   exponentiate = T  
8 ) |>  
9 add_global_p()
```

Characteristic	HR ¹	95% CI ¹	p-value
Treatment			<0.001
Obs	—	—	
Lev	0.99	0.80, 1.22	
Lev+5FU	0.60	0.47, 0.75	
Sex			0.3
Female	—	—	
Male	0.90	0.75, 1.08	
¹ HR = Hazard Ratio, CI = Confidence Interval			

Time-to-event plots with {ggsurvfit}

Uses {ggplot2} as the basis so known customizations are available with the `+` operator.

- Plot Kaplan-Meier curves using `ggsurvfit`
- Plot cumulative incidence curves for competing risks using `ggcumin`
- Plot multi-state models using `ggsurvfit`

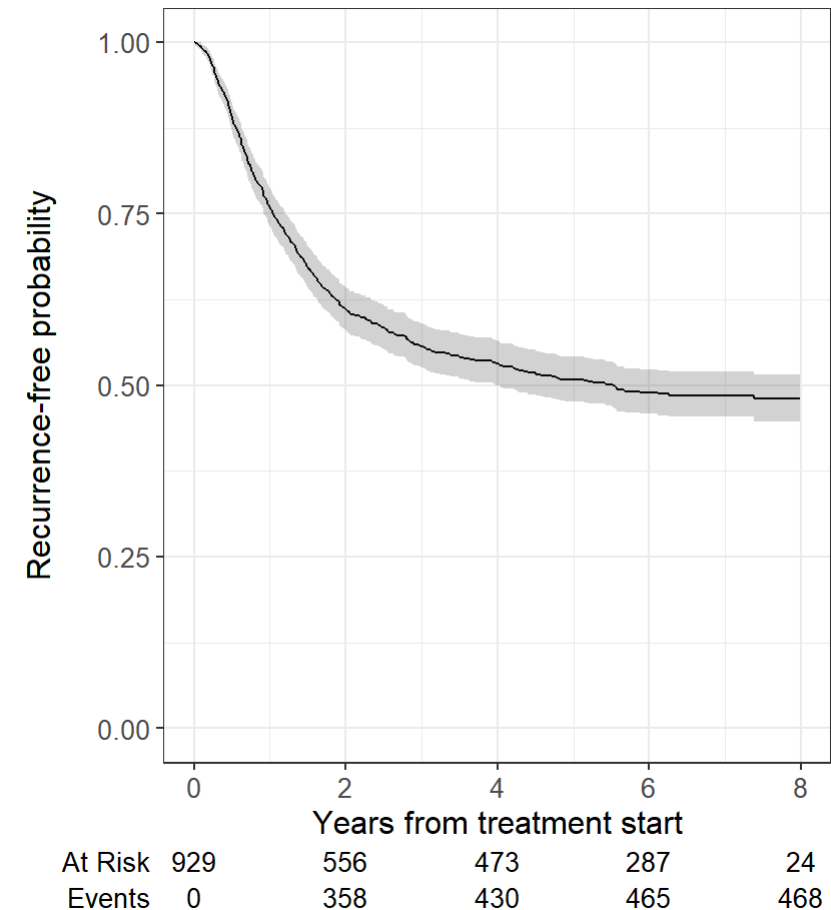
Options to:

- Add confidence intervals
- Add risk tables
- Add quantiles
- And more

See <https://www.danielsjoberg.com/ggsurvfit/> for details and vignettes to get started

Kaplan-Meier curves with ggsurvfit()

```
1 survfit2(  
2   Surv(years, status) ~ 1,  
3   data = colonr  
4 ) |>  
5 ggsurvfit() +  
6 add_confidence_interval() +  
7 add_risktable() +  
8 scale_y_continuous(  
9   limits = c(0, 1)  
10 ) +  
11 scale_x_continuous(  
12   limits = c(0, 8),  
13   breaks = seq(0, 8, 2)  
14 ) +  
15 labs(  
16   x = "Years from treatment start",  
17   y = "Recurrence-free probability",  
18 )
```



Put it all together

```

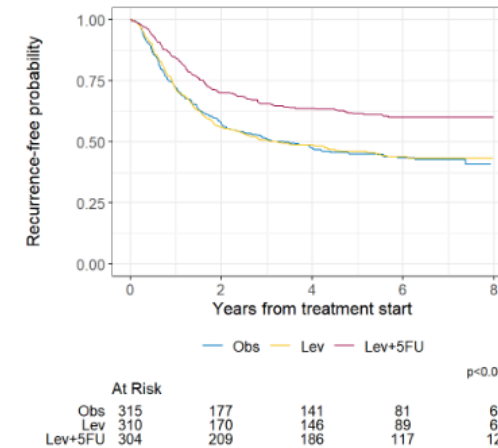
1 ---
2 title: "Colon recurrence report"
3 format: html
4 execute:
5   echo: false
6   warning: false
7   error: false
8 ---
9
10 {r}
16
17 {r}
34
35 In a clinical trial, patients were randomized to receive adjuvant chemotherapy with either
    levamisole, levamisole + 5-FU, or placebo. Interest was in whether treatment group was
    associated with time to recurrence.
36
37 Time to recurrence was measured from the date of treatment start to the date of recurrence
    or last known date the patient was recurrence-free. The Kaplan-Meier method was used to
    estimate recurrence-free probabilities. Multivariable Cox regression adjusted for age and
    sex.
38
39 {r fig.width = 5, fig.height = 4.5}
40 survfit2(Surv(years, status) ~ rx, data = colonr) |>
41   ggsurvfit() +
42   add_risktable(risktable_stats = "n.risk") +
43   scale_y_continuous(limits = c(0, 1)) +
44   scale_x_continuous(limits = c(0, 8), breaks = seq(0, 8, 2)) +
45   scale_color_manual(values = ezfun::ccf_palette("contrast")) +
46   labs(
47     x = "Years from treatment start",
48     y = "Recurrence-free probability"
49   ) +
50   add_pvalue()
51
52
53 {r}
54 tab1 <-
55   colonr |>
56   tbl_survfit(
57     y = Surv(years, status),
58     include = rx,
59     times = 5,
60     label_header = "{time}-year RFS (95% CI)"
61   )
62
63 tab1
64
65
66 The 5-year (95% CI) recurrence-free probability is `r inline_text(tab1, variable = rx,
    level = 'obs', time = 5)`, `r inline_text(tab1, variable = rx, level = 'Lev', time = 5)`,
    and `r inline_text(tab1, variable = rx, level = 'Lev+5FU', time = 5)` for the placebo,
    levamisole, and levamisole + 5-FU groups, respectively.
67
68 {r}
69 tab2 <-
70   coxph(Surv(years, status) ~ rx + sex + age, data = colonr) |>
71   tbl_regression(exponentiate = T) |>
72   add_global_p()
73
74 tab2
75
76
77 Levamisole + 5-FU was associated with a hazard ratio of `r inline_text(tab2, variable =
    rx, level = 'Lev+5FU', pattern = "{estimate} (95% CI {conf.low}, {conf.high})"` for
    recurrence as compared to placebo.

```

Colon recurrence report

In a clinical trial, patients were randomized to receive adjuvant chemotherapy with either levamisole, levamisole + 5-FU, or placebo. Interest was in whether treatment group was associated with time to recurrence.

Time to recurrence was measured from the date of treatment start to the date of recurrence or last known date the patient was recurrence-free. The Kaplan-Meier method was used to estimate recurrence-free probabilities. Multivariable Cox regression adjusted for age and sex.



Characteristic 5-year RFS (95% CI)

Treatment

Obs	45% (40%, 51%)
Lev	46% (41%, 52%)
Lev+5FU	62% (56%, 67%)

The 5-year (95% CI) recurrence-free probability is 45% (40%, 51%), 46% (41%, 52%), and 62% (56%, 67%) for the placebo, levamisole, and levamisole + 5-FU groups, respectively.

Characteristic	HR [†]	95% CI [†]	p-value
Treatment			<0.001
Obs	—	—	
Lev	0.99	0.80, 1.22	
Lev+5FU	0.60	0.48, 0.76	
Sex			0.3
Female	—	—	
Male	0.90	0.75, 1.08	
age	0.99	0.99, 1.00	0.11

[†] HR = Hazard Ratio, CI = Confidence Interval

Thank you, and please get in touch

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Link to slides:

<https://github.com/zabore/slidedecks/blob/master/2024-NYR-Zabor.html>

Appendix: Create example dataset

- `colon` dataset from `{survival}` altered to:
 - Limit to one row per patient with interest in recurrence rather than death
 - Scale time from days to years
 - Recode sex levels to informative values
 - Label variables of interest

```
1 colonr <-  
2   colon |>  
3   filter(etype == 1) |>  
4   mutate(  
5     years = time / 365.25,  
6     sex = case_match(  
7       sex,  
8       0 ~ "Female",  
9       1 ~ "Male"  
10    )  
11  ) |>  
12  labelled::set_variable_labels(  
13    rx = "Treatment",  
14    sex = "Sex"  
15  )
```